Content Improvement Ideas for Chapter 4

Application to AR/ARMA Processes - thesis development report

The chapter contains two particularly promising ideas: the prediction-based dimension lift and the geometric interpretation through zonotopes. The main opportunity is to turn the current collection of exploratory ideas into one coherent statistical argument with a precise target, a population justification, a fully specified estimator, and a convincing numerical evaluation.

1. Sharpen the chapter's central question

State one precise, testable objective near the beginning. A suitable formulation is: *Can the autoregressive order p be identified as the smallest embedding dimension k after which out-of-sample prediction geometry stabilizes?* Organize the subsequent constructions around this question. At present, scale selection, simplicial complexes, covariance, zonotopes, intrinsic volumes, and minimum-volume sets compete for attention.

2. Decide whether the chapter concerns AR or ARMA models

The title refers to ARMA processes, while almost all substantive results concern AR processes. Either rename the chapter *Application to Autoregressive Processes*, or add a genuine ARMA extension and explain whether the selected scale is intended to estimate p, q, or both. Restricting the chapter to AR models would create the cleaner and more defensible thesis.

3. Add the key theoretical proposition

Turn the current intuition - prediction is poor for k < p and stabilizes for k >= p - into a formal population result. Under suitable non-degeneracy assumptions, show that the optimal one-step linear prediction risk has a strict improvement up to the true order and no further population improvement afterward. This would provide the statistical foundation for identifying p as a critical scale.

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R_k > R_p = innovation variance for k < p; R_k = innovation variance for k >= p.
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4. Resolve the heavy-tail contradiction

The chapter aims to avoid requiring second moments, but its main lift uses ordinary least squares, covariance matrices, and averages of products X_t X_s. These can fail for heavy-tailed innovations. Present two explicit variants: a finite-variance version using OLS and covariance, and a robust version using least absolute deviations, Huber loss, quantile regression, rank dependence, or bounded transforms. If F_0(X_t) is used, note that F_0 is unknown and must be replaced by empirical ranks or an empirical CDF.

5. Clarify the role of Takens' theorem

Takens' theorem concerns deterministic dynamical systems and generic observation maps; it does not directly justify delay embeddings of stochastic AR processes. Present it as historical motivation or analogy. The direct justification in this chapter should instead come from the finite-memory property of AR models and their linear-process representations.

6. Prevent in-sample overfitting

The completion matrix is fitted from the observations and the proposed prediction errors appear to be assessed on the same observations. Since flexibility increases with k, this can create artificial improvement. Use a chronological training/test split, rolling-origin evaluation, or blocked cross-validation. Construct the scale statistics and complexes from out-of-sample residuals.

7. Specify one complete data-dependent complex

Choose one primary construction rather than listing several possibilities. For example, calculate out-of-sample completion residuals and use their pairwise differences as edge weights. Then define a threshold graph or nearest-neighbour graph and take its flag complex. Explain how the threshold is selected; a quantile-based rule would improve comparability across embedding dimensions.

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e_t^(k) = z_t^(k) - A_hat_k Y_t^(k); w_k(s,t) = ||e_s^(k) - e_t^(k)||.
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8. Reconsider the temporal simplex construction

The initial complex based only on consecutive time indices is independent of the observed process, so much of its topology may be predetermined. Explain what information remains after the data-dependent restriction. A recurrence graph on lag vectors or residual vectors may give a more direct connection between geometry, temporal recurrence, and persistent homology.

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Connect s and t when ||Y_s^(k) - Y_t^(k)|| <= epsilon.
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9. Strengthen and correct technical statements

Several statements should be refined: the condition sum |phi_i| < 1 is sufficient, not necessary, for causality; centered innovations imply a zero mean signed distance from the AR hyperplane but not concentration near it; for k > p+1, slab intersection gives containment unless equality is proved; and the Gaussian ellipsoid threshold can be expressed through a chi-square quantile. For the truncated zonotope, add a Hausdorff error bound controlled by the omitted generator norms.

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d_H(Z, Z_n) <= sum_{j>n} ||g_j||.
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10. Make the numerical study substantially more informative

Vary the true order, sample size, coefficient configuration, distance from the unit-root boundary, innovation distribution, and innovation scale over many repetitions. Compare the proposed method with AIC, BIC, the PACF cut-off, cross-validated prediction, and robust order-selection procedures. Report selection accuracy, overestimation and underestimation rates, stability, and runtime rather than relying on individual plots.

11. Add negative controls

Include white noise, moving-average processes, nonlinear time series, and nonstationary or near-unit-root processes. These examples test whether the method recognizes cases without a clear finite AR order, rather than producing a peak for every dataset.

12. Restructure the chapter

A clearer progression is: (1) research objective and estimand; (2) AR(p) assumptions; (3) a population prediction result identifying p; (4) finite-sample and robust estimators; (5) complex construction and scale statistic; (6) geometric interpretations for uniform and Gaussian innovations; (7) simulation design; (8) results and classical comparisons; and (9) limitations and a possible ARMA extension.

Highest-priority recommendation: prove that prediction risk stabilizes at k = p, and then present the geometric or topological construction as a method for detecting that stabilization. This single bridge would make the chapter read as a unified statistical method rather than a collection of interesting constructions.